

1. Registry (The Heart of Windows)

The Windows Registry stores configuration settings and activity data. Key hives to monitor:

HKEY_CURRENT_USER (HKCU) and HKEY_LOCAL_MACHINE (HKLM)

- **User Activity and Persistence:**
 - HKCU\Software\Microsoft\Windows\CurrentVersion\Run: Programs that execute at user login (possible persistence mechanism).
 - HKLM\SYSTEM\CurrentControlSet\Services: Installed services (can be used for persistence).
 - HKCU\Software\Microsoft\Windows\Shell\BagMRU: Tracks folder access.
 - **USB Device History:**
 - HKLM\SYSTEM\CurrentControlSet\Enum\USBSTOR: Lists connected USB devices.
 - **User Last Login:**
 - HKCU\Volatile Environment: Tracks logon sessions.
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2. Prefetch Files (C:\Windows\Prefetch)

- **Purpose:** Tracks recently executed programs for faster access.
 - **Forensics Value:**
 - Identifies programs executed (including malware).
 - Contains timestamps of execution.
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3. Event Logs (C:\Windows\System32\winevt\Logs)

Windows Event Logs record detailed system activities. Monitor key logs:

- **System Logs (System.evtx):** System events like shutdowns, reboots, and driver failures.
 - **Security Logs (Security.evtx):** Logon and logoff events, privilege escalations.
 - Event ID **4624**: Successful logon.
 - Event ID **4625**: Failed logon attempt.
 - Event ID **4688**: Process creation (useful for tracking malware).
 - **Application Logs (Application.evtx):** Logs generated by installed software.
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4. Windows Shimcache (AppCompatCache)

- Tracks programs that have been executed on the system.
 - Located in the **Registry at** `HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\AppCompatCache`.
 - Useful for identifying programs executed, even if the binary has been deleted.
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5. User Internet Activity (Web Browsing)

- **Internet Explorer/Edge:**
 - `C:\Users\%Username%\AppData\Local\Microsoft\Windows\History:` Browsing history.
 - `WebCacheV01.dat`: Stores history, cookies, and cache.
 - **Chrome/Firefox:**
 - `C:\Users\%Username%\AppData\Local\Google\Chrome\User Data\Default\History`: SQLite database containing browsing history.
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6. Recycle Bin (`C:\$Recycle.Bin`)

- **Forensics Value:** Contains deleted files that may still be recoverable, including timestamps.
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7. Recent Files and Jump Lists

- **RecentDocs** (`C:\Users\%Username%\AppData\Roaming\Microsoft\Windows\Recent`): Tracks recently accessed files.
 - **Jump Lists** (`%AppData%\Microsoft\Windows\Recent\AutomaticDestinations`): Stores info about recently opened files and programs.
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8. Pagefile and Hibernation File (`pagefile.sys` and `hiberfil.sys`)

- **Purpose:** Virtual memory and system state storage.
 - **Forensics Value:** Contains fragments of data, passwords, and session details.
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9. Shadow Copies and System Restore Points (C:\System Volume Information)

- **Forensics Value:** Historical snapshots of the system state. Can recover deleted files and previous Registry states.
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10. Windows Task Scheduler (C:\Windows\System32\Tasks)

- Monitor for malicious persistence mechanisms:
 - Scheduled tasks created by attackers for persistence or privilege escalation.
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11. LNK Files (Shortcuts)

- Found in C:\Users\%Username%\Recent.
 - Reveal file paths and timestamps, even if the original file is deleted.
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12. Windows Logs for Remote Desktop Connections

- C:\Windows\System32\winevt\Logs\Microsoft-Windows-TerminalServices-*
 - Tracks remote desktop connections (successful and failed attempts).
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13. BitLocker Logs (C:\Windows\System32\LogFiles)

- Useful in investigating disk encryption attempts.